Exercise sheet: Week 4 - solutions

1. We have

$$\widehat{\lambda} = \frac{1}{\overline{X}}, \quad \text{CRLB} = \frac{\lambda^2}{n}, \quad \widehat{\lambda} \approx \text{N}\left(\lambda, \frac{\lambda^2}{n}\right), \text{ and so } \frac{\widehat{\lambda} - \lambda}{\sqrt{\frac{\lambda^2}{n}}} \approx \text{N}(0,1)\,.$$

Therefore, the 95% CI for λ is $\hat{\lambda}\left(1 \pm \frac{1.96}{\sqrt{n}}\right)^{-1}$.

Data: $\hat{\lambda} = 0.578$, with CI given by $(0.357, 1.520)$.

(a) We write $\theta = \frac{1}{\lambda}$. Then, it follows that the MLE is $\widehat{\theta} = \overline{X}$, and $\text{CRLB} = \frac{\theta^2}{n}$. Thus, we obtain

$$\frac{\widehat{\theta} - \theta}{\sqrt{\frac{\theta^2}{n}}} \approx \text{N}(0,1)$$

and that the required CI is $\hat{\theta}\left(1 \pm \frac{1.96}{\sqrt{n}}\right)^{-1}$. Data

$$n = 10, \quad \overline{x} = 1.73, \quad \widehat{\theta} = \overline{x} = 1.73 = \frac{1}{0.578} = \frac{1}{\widehat{\lambda}}.$$

The MLEs agree (by the invariance property).

For the CI we insert $\hat{\theta}$ and n to obtain

$$\left(\frac{1.73}{1 + \frac{1.96}{\sqrt{10}}}, \frac{1.73}{1 - \frac{1.96}{\sqrt{10}}}\right) = (1.068, 4.55)\,.$$

Converting to $\lambda = \frac{1}{\theta}$ gives a CI

$$\left(\frac{1}{4.55}, \frac{1}{1.068}\right) = (0.22, 0.936).$$

Hence, *the CIs do not agree*. Note that if the double approximation is used, the confidence intervals do agree.

(b) It is obvious that the CIs agree in this case. Now consider the log likelihood *for both parametrisations*. For λ and θ we obtain, respectively,

$$\lambda: \quad l_1(\lambda) = n\,\log(\lambda) - \lambda \sum_{i=1}^{10} x_i,$$

$$\theta: \quad l_2(\theta) = -n\,\log(\theta) - \frac{1}{\theta}\sum_{i=1}^{10} x_i\,,$$

and so

$$l_1(\lambda) = l_2\left(\frac{1}{\lambda}\right).$$

Therefore, $\widehat{\lambda}$ maximises l_1 if and only if $\widehat{\theta}$ maximises l_2.

Consider the *deviances*

$$D_1(\lambda) = 2\big(l_1(\widehat{\lambda}) - l_1(\lambda)\big),$$
$$D_2(\theta) = 2\big(l_2(\widehat{\theta}) - l_2(\theta)\big).$$

Note that $l_1(\widehat{\lambda}) = l_2(\widehat{\theta})$.

Let I_λ, I_θ be the two CIs obtained for λ and θ, respectively. Then we have

$$\begin{aligned}
\lambda \in I_\lambda &\Leftrightarrow D_1(\lambda) < 3.841 \\
&\Leftrightarrow 2\left(l_1(\widehat{\lambda}) - l_1(\lambda)\right) < 3.841 \\
&\Leftrightarrow 2\left(l_2\left(\frac{1}{\widehat{\lambda}}\right) - l_2\left(\frac{1}{\lambda}\right)\right) < 3.841 \\
&\Leftrightarrow \theta = \frac{1}{\lambda} \in I_\theta.
\end{aligned}$$

This shows that the CIs agree.

2. (a) The Poisson(μ) distribution has pmf

$$f_x(x;\mu) = \frac{\mu^x\, e^{-\mu}}{x!}$$

and we have

$$l(\mu) = \text{constant} + n\,\overline{x}\,\log(\mu) - n\,\mu,$$

which implies that $l'(\mu) = \frac{n\,\overline{x}}{\mu} - n$ and the MLE is $\widehat{\mu} = \overline{x}$.

We compute

$$\begin{aligned}
l(\widehat{\mu}) &= \text{constant} + n\,\overline{x}\,\log(\overline{x}) - n\,\overline{x} \\
\Rightarrow D(\mu) &= 2(l(\widehat{\mu}) - l(\mu)) \\
&= 2n\,(\overline{x}\,\log(\overline{x}) - \overline{x}\,\log\mu + \mu - \bar{x}) \\
&= 2n\,\left((\mu - \overline{x}) + \overline{x}\,\log\left(\frac{\overline{x}}{\mu}\right)\right),
\end{aligned}$$

as required.

(b) For $n = 100$, $\sum_i x_i = 74$, we have $\bar{x} = 0.74$. The 99%-CI is

$$\left\{\mu\,:\,D(\mu) < 6.635 = \chi^2_1(0.01)\right\}.$$

To obtain this using R, one can do the following:

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thresh = qchisq(0.99,1)
xbar = 0.74
n = 100
mu = c(1:1000)*0.0005 + 0.5
dev = 2*n*(mu - xbar + xbar*log(xbar/mu))
ci = mu[dev<thresh]
ci[1]
ci[length(ci)]
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This gives

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ci[1] = 0.54
ci[length(ci)] = 0.984,
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i.e., the confidence interval $(0.54, 0.98)$.

3. (a) We have that

$$U(\mu; \underline{X}) = \frac{n}{\sigma^2}(\overline{X} - \mu) \quad \text{and} \quad I(\mu) = \frac{n}{\sigma^2}.$$

Note that $I(\mu)$ is a constant independent of μ. Hence,

$$I(\widehat{\mu}) = I(\mu) = \frac{n}{\sigma^2}.$$

The asymptotic normality of the MLE states that (subject to regularity conditions)

$$(\overline{X} - \mu)\sqrt{I(\mu)} \approx \mathrm{N}(0, 1),$$

but we know that $\overline{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$. Therefore,

$$(\bar{X} - \mu)\sqrt{I(\mu)} \sim \mathrm{N}(0, 1)$$

exactly for all n. The asymptotic normality property here holds exactly for all n (so also non-asymptotically), using both $I(\hat{\mu})$ and $I(\mu)$.

(b) We compute

$$\begin{aligned}
D(\mu) &= 2\,(l(\widehat{\mu}) - l(\mu)) \\
&= 2\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \overline{x})^2 + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right) \\
&= \frac{1}{\sigma^2}\left(\left(\sum_{i=1}^{n} x_i^2 - 2\mu\, n\,\overline{x} + n\,\overline{x}^2\right) - \left(\sum_{i=1}^{n} x_i^2 - n\,\mu^2\right)\right) \\
&= \frac{n}{\sigma^2}\left(\overline{x}^2 - 2\mu\,\overline{x} + \mu^2\right) \\
&= \frac{n}{\sigma^2}(\bar{x} - \mu)^2.
\end{aligned}$$

For the sampling distribution we have $\overline{X} \sim \text{N}(\mu, \frac{\sigma^2}{n})$, so that

$$D(\mu) = \left(\frac{\overline{X} - \mu}{\sqrt{\frac{\sigma^2}{n}}} \right)^2 \sim \chi^2_1.$$

So the asymptotic distribution of the deviance holds exactly *for all* n.

4. (a) Using $\frac{\overline{X}-\theta}{\sqrt{\theta/n}} \approx \text{N}(0,1)$ as pivot:

$$\mathbf{P}\left(-Z_{\alpha/2} < \frac{(\overline{X} - \theta)}{\sqrt{\theta/n}} < Z_{\alpha/2} \right) = 1 - \alpha$$
$$\iff \mathbf{P}\left(\frac{(\overline{X} - \theta)^2}{\theta/n} < Z^2_{\alpha/2} \right) = 1 - \alpha$$

We need to find the range of θ for which this is true. To identify this range we rearrange to give the quadratic inequality

$$\theta^2 - \left(2\overline{X} + \frac{Z^2_{\alpha/2}}{n} \right) \theta + \overline{X}^2 < 0\,.$$

The CI (θ_L, θ_U) comes from the roots, i.e.

$$\theta_L, \theta_U = \overline{X} + \frac{Z^2_{\alpha/2}}{2n} \pm Z_{\alpha/2} \sqrt{\frac{\overline{X}}{n} + \frac{Z^2_{\alpha/2}}{4n^2}}\,.$$

(b) It is much easier to use $\frac{\overline{X}-\theta}{\sqrt{\widehat{\theta}/n}} \approx \text{N}(0,1)$, where $\widehat{\theta} = \overline{X}$. This gives the CI $\overline{X} \pm Z_{\alpha/2}\sqrt{\frac{\overline{X}}{n}}$.

Data: $n = 20$, $\sum x_i = 94$, and so $\bar{x} = 4.7$

(a) $4.7 + \frac{1.96^2}{40} \pm 1.96\sqrt{\frac{4.7}{20} + \frac{1.96^2}{4\cdot(20)^2}}$, or 4.796 ± 0.955, or $(3.84, 5.75)$.

(b) $4.7 \pm 1.96\sqrt{\frac{4.7}{20}}$, or 4.7 ± 0.95, or $(3.75, 5.65)$.

CIs are similar, with roughly the same width, but (a) is centred higher.